

# DAYANANDA SAGAR UNIVERSITY, BENGALURU

School of Engineering • End Semester Examination

Course: Signals and Systems | Course Code: 21EC42 | Credits: 4

Exam Session: End Semester Exam 2024

Max Marks: 80 | Duration: 3 Hours

## INSTRUCTIONS TO CANDIDATES:

1. Answer FIVE full questions, choosing ONE full question from each module.
2. Draw neat diagrams wherever necessary. Missing data may be suitably assumed.

## MODULE 1

- Q1.** Evaluate the continuous-time convolution integral  $y(t) = x(t) * h(t)$  where  $x(t) = e^{-2t} u(t)$  and  $h(t) = u(t - 1)$ . Plot the output signal  $y(t)$ .

[8 Marks]